

Supplementary Online Content

DeBar LL, Mayhew M, Wellman RD, et al. Telehealth and online cognitive behavioral therapy–based treatments for high-impact chronic pain: a randomized clinical trial. *JAMA*. Published online July 23, 2025. doi:10.1001/jama.2025.11178

OUTCOME MEASURES	2
INTERVENTION ADHERENCE	3
MISSING DATA	4
Adjustment:	4
Imputation (MNAR):	4
Non-Response Weighting:	5
SENSITIVITY ANALYSES	11
Missing Data Analyses:	11
Protocol Change Subset Analyses:	13
ADDITIONAL UNADJUSTED DATA AND ANALYSES	15
eREFERENCES	19

eFigure 1. Histogram of Number of Sessions completed for those randomized to painTRAINER and those randomized to the Virtual Health Coach	3
eTable 1. Primary outcome follow-up rates and patterns overall and by arm.	7
eTable 2. Baseline variables used in process of selecting covariates related to missingness and to MCID of pain severity at 3 months.	7
eTable 3. Imputation step 1: Summary of fitted imputation model for primary analysis.	8
eTable 4. Imputation step 2: Summary of primary estimation model fit to imputed data with weighting. ..	9
eTable 5. Summary of logistic weighting model for any observed follow-up.	10
eTable 6. Summary of estimated weights.	10
eFigure 2. Histogram of estimated probabilities and weights.	10
eTable 7. Pre-specified Sensitivity Analyses for different approaches to handling missing data.	11
eTable 8. Results of a Subset of Participants after two different protocol changes: Enhancement to painTRAINER (randomized after 8/9/2021) and Adding follow-up letters to improve follow-up rates (randomized after 3/31/2022)	14
eTable 9. Unadjusted percentage of participants with clinically meaningful improvement by treatment group and adjusted relative risk between group comparisons for the binary primary outcome (pain severity) and secondary pain outcomes	15
eTable 10. Unadjusted Median and Interquartile Ranges for change from baseline of continuous secondary outcomes by treatment group and unadjusted mean differences between group comparisons	16
eTable 11. Unadjusted Median and Interquartile Ranges of continuous secondary outcomes by treatment group and unadjusted mean differences in between group comparisons	17

This supplementary material has been provided by the authors to give readers additional information about their work.

OUTCOME MEASURES

Pain Severity Score: Based on modified 11-item version of the Brief Pain Inventory – Short Form (BPI-SF). Score is calculated mean of all 11 items; range 0-10 with a higher score = worse pain severity

Pain Intensity Score: Based on 4-item subscale of the Brief Pain Inventory – Short Form (BPI-SF). Score is calculated mean of all 4 items; range 0-10 with a higher score = worse pain intensity

Pain-related Interference Score: Based on 7-item subscale of the Brief Pain Inventory – Short Form (BPI-SF). Score is calculated mean of 7 items; range 0-10 with a higher score = worse pain-related interference

Minimal clinically important difference (MCID) is defined as $\geq 30\%$ decrease in pain score from baseline (i.e., improvement; pain score range 0-10 with a higher score = worse pain severity, intensity or related interference)

8-Item Patient Health Questionnaire Depression Scale (PHQ-8): Sum of 8 items related to depressive symptoms (each item response 0-3) with total scale range=0-24; higher score = greater severity of depressive symptoms. Cut points= 10 (moderate), 15 (moderately severe) and 20 (severe depression).

The Generalized Anxiety Disorder 7-item Scale (GAD-7): Sum of 7 items related to anxiety symptoms (each item response 0-3) with total scale range=0-21; higher score = greater severity of anxiety symptoms. Cut points= 5 (mild), 10 (moderate) and 15 (severe) levels of anxiety.

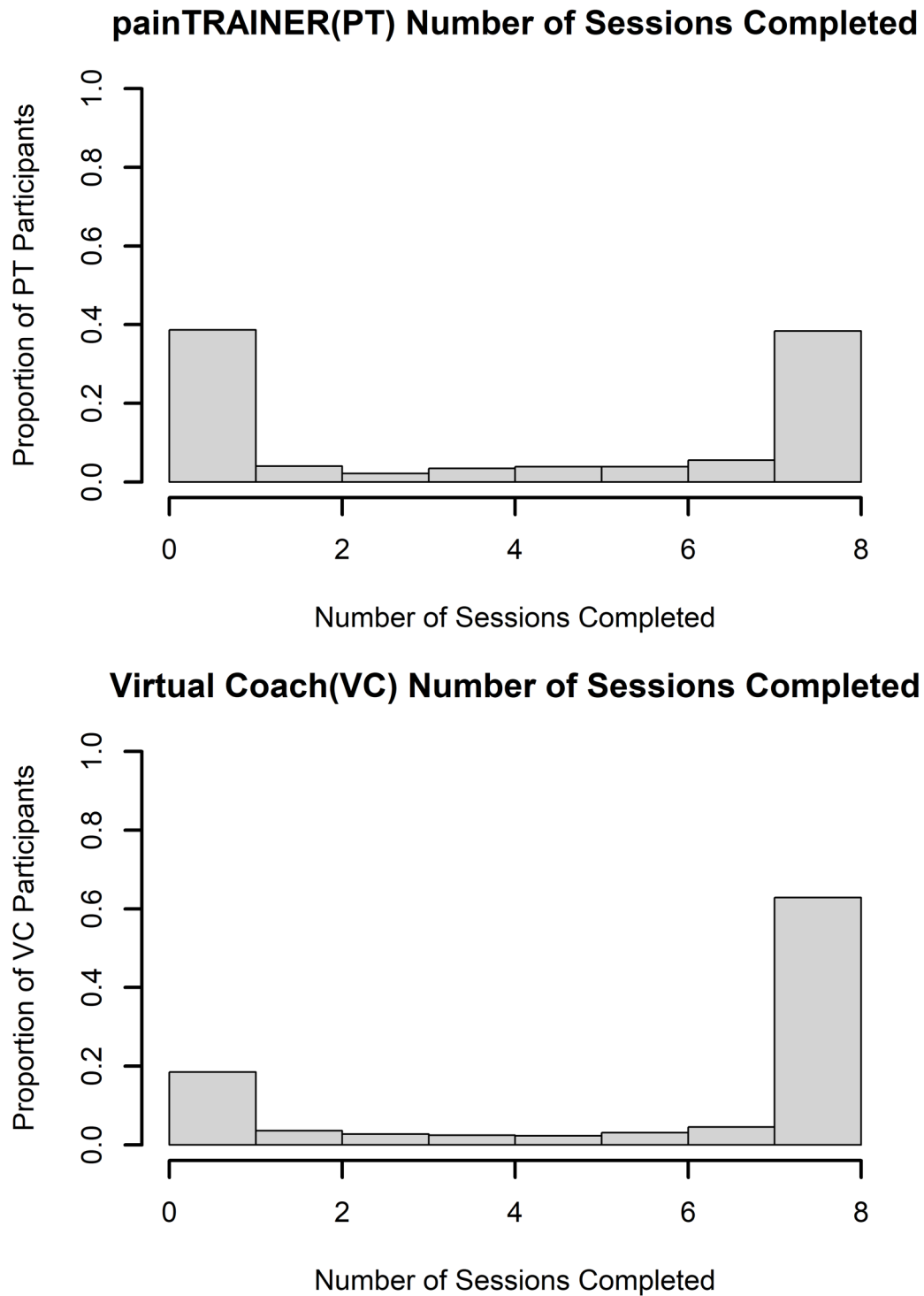
PROMIS (Patient-Reported Outcomes Measurement Information System) Sleep Disturbance – Short Form 6a: Score by summing the 6 items (each item response 1-5) and then converting raw score to standardized T-score, using the HealthMeasures Scoring Service, with mean of 50 and standard deviation of 10. Higher T-score= worse sleep disturbance. Score ≥ 60 indicates moderate (60-70) to severe (>70) sleep disturbance.

PROMIS Ability to Participate in Social Roles and Activities – Short Form 4a: Score by summing the 4 items (each item response 1-5) and then converting raw score to standardized T-score, using the HealthMeasures Scoring Service, with mean of 50 and standard deviation of 10. Higher T-score= better ability to participate in social roles and activities. Score ≤ 40 indicates moderate (40-30) to severe (< 30) limitations in ability to participate.

PROMIS Physical Function – Short Form 6b: Score by summing the 6 items (each item response 1-5) and then converting raw score to standardized T-score, using the HealthMeasures Scoring Service, with mean of 50 and standard deviation of 10. Higher T-score= better physical functioning. Score ≤ 40 indicates moderate (40-30) to severe (< 30) limitations in physical functioning.

INTERVENTION ADHERENCE

eFigure 1. Histogram of Number of Sessions completed for those randomized to painTRAINER and those randomized to the Virtual Health Coach



MISSING DATA

The study met the pre-specified criteria enumerated in the statistical analysis plan (SAP; **Supplement 2**) for the primary analysis to use an imputation approach which assumes the data are missing not at random (MNAR) in combination with non-response weighting and regression adjustment. The specific criteria were missingness of the primary outcome, MCID in pain severity (30% reduction from baseline), at 3-month (primary time-point) above 15% or differential missingness between arms. As shown in **eTable 1**, the 3-month primary outcome was complete 77.1% overall (79.9% UC, 69.8% PT, and 81.6% HC). Overall, 67.3% of individuals had all three time points observed and 12.7% of individuals were missing follow-up at all three time points.

Details regarding the imputation procedure, the models for imputation and for the weights, measured covariates, and variable selection are provided in the sections below. **Supplement 5** displays the distributions of all measured baseline characteristics, both overall and within groups defined by three patterns of observed follow-up data for the primary outcome: 1) no recorded follow-up measures, 2) one or 2 follow-up measures recorded, or 3) all follow-up measures recorded.

Adjustment: The statistical analysis plan for the primary outcome pre-specified adjustment for baseline levels of pain severity, other stratification variables (sex, clinical site, and rural/medically underserved residency), and a priori variables predictive of outcome (multisite pain and co-occurring mental health condition). Additional predictor variables were assessed and included in the imputation and estimation models if they were related both to missingness of the 3-month time-point and were predictive of MCID in pain severity at 3 months. Specific additional predictors for the imputation and estimation models were selected after assessing all other available covariates, one-by-one, in separate logistic regression models for missingness at 3-months, and in separate modified Poisson models¹ for MCID in pain severity at 3-months, while adjusting for study arm and the base set of pre-specified adjustment variables: baseline pain severity, age, sex, clinical site, rural/medically underserved residency, multisite pain, and co-occurring mental health conditions. In this context, if a covariate was significant at the 0.10 level in both the missingness and outcome models while adjusting for the base set of predictors, then it was added as an adjustment factor in the imputation and estimation models for the main analysis. A similar process was used to select additional covariates for the weight model, details of which are described below in the section on non-response weighting. **eTable 2** provides a list of variables that were assessed during the selection process and provides a list of additional variables that were selected for the main models and the weight model. **Supplement 5** also provides p-values for variables that were significant in one or more of the selection models used to identify additional adjustment variables, i.e., variables that were significantly ($P < 0.10$) related to the binary outcome MCID in pain severity at 3 months and/or the presence of missing values in the 3-month pain severity, and/or the likelihood of missing all follow-up points (non-response).

Imputation (MNAR): Implementation of the MNAR imputation approach for GEE² requires the specification and execution of a two-step modeling process. The first step is to impute missing outcome data using a pattern-mixture approach with reasonable choices for included missing patterns given the amount of missing data. To classify patterns of missing data in this study, among those with any observed follow-up measure we created a categorical variable indicating whether each participant had one (8.1%), two (12.1 %) or three (67.3%) follow-up observations recorded. The full imputation model was specified as follows:

$$\begin{aligned} E(Y_{im}|X_i, \mathbf{Z}_i) = & \alpha + \beta_{PT}I(X_i = PT) + \beta_{HC}I(X_i = HC) + \beta_z\mathbf{Z}_i + \beta_6I(m = 6) + \beta_{12}I(m = 12) \\ & + \gamma_{PT,6}I(X_i = PT)I(m = 6) + \gamma_{PT,12}I(X_i = PT)I(m = 12) \\ & + \gamma_{HC,6}I(X_i = HC)I(m = 6) + \gamma_{HC,12}I(X_i = HC)I(m = 12) \\ & + \eta_1I(Ob_{si} = 1) + \eta_2I(Ob_{si} = 2) \\ & + \eta_{PT,1}I(X_i = PT)I(Ob_{si} = 1) + \eta_{HC,1}I(X_i = HC)I(Ob_{si} = 1) \\ & + \eta_{PT,2}I(X_i = PT)I(Ob_{si} = 2) + \eta_{HC,2}I(X_i = HC)I(Ob_{si} = 2) \end{aligned}$$

where Y_{im} is the change in pain severity for individual i at time point m , X_i is the intervention group (UC, PT, and UC), m is the number of months post randomization, and $\mathbf{Z}_i$ is the vector of baseline covariates. Further, Obs_i is a variable that indicates the pattern of observed outcomes present. In this case, Obs_i takes on the values 1, 2 or 3 depending on whether 1, 2 or 3 of the follow-up measures are observed for a randomized person. The $\mathbf{Z}_i$ include all adjustment variables as specified previously. We fit this assuming a modified linear model using GEE with independent working correlation and robust standard errors, accounting for clustering of participants within health coaches and multiple observations per person. The fitted imputation model was then used to replace missing outcomes data with predicted values from the model. The binary MCID outcome variable was then calculated using the change outcome with imputed data and the baseline value of pain severity.

In the second step, the following estimation model was fit to the full imputed data with the binary MCID outcome now represented as $\hat{Y}_{im}$ and using the following pre-specified functional form from the statistical analysis plan:

$$\begin{aligned} \log(E(\hat{Y}_{im}|X_i, \mathbf{Z}_i)) = & \alpha + \beta_{PT}I(X_i = PT) + \beta_{HC}I(X_i = HC) + \beta_z\mathbf{Z}_i + \beta_6I(m = 6) + \beta_{12}I(m = 12) \\ & + \gamma_{PT,6}I(X_i = PT)I(m = 6) + \gamma_{PT,12}I(X_i = PT)I(m = 12) \\ & + \gamma_{HC,6}I(X_i = HC)I(m = 6) + \gamma_{HC,12}I(X_i = HC)I(m = 12). \end{aligned}$$

The model was fit with GEE assuming Poisson errors and using robust sandwich standard errors to account for clustering within provider and individual.¹ Parameter estimates are taken directly from the solution to the second step GEE model above and the variance-covariance matrix for the joint imputation and analysis model are calculated as described in Wang and Fitzmaurice.² Summaries of the fitted imputation and estimation models are provided in **eTable 3** and **eTable 4**.

As a final note on imputation, there were a small number of cases where baseline covariates contained a limited number of missing values. To keep the entire randomized cohort intact for all analyses, these missing baseline values were imputed with their corresponding population means estimated from all available baseline data.³

Non-Response Weighting: The imputation approach assuming data are MNAR was used to impute missing data for those that had at least one observation, but we observed a higher-than-expected number of participants with all outcome time points missing (12.7%) and a differential by study arm: 9.5% for Usual care plus, 17.1% for painTRAINER, and 11.3% for health coach (**eTable 1**). To be able to estimate an average treatment effect (ATE) amongst all randomized, and due to concern about differential missingness, we added non-response weighting which we formally incorporated into the SAP.

For the primary weighted analysis, to estimate the non-response weights, we fit a logistic regression model to the entire randomized population, with an indicator of at least one observed follow-up as the outcome. Predictor variables included study arm, baseline pain severity, age, sex, clinical site, rural/medically underserved residency, multisite pain, and co-occurring mental health conditions and additional variables that were associated both with the 3-month MCID in pain severity outcome and with having any follow-up at 3, 6 or 12 months (**Supplement 5**). As detailed above for adjustment variables, additional variables for the weight model were evaluated one-at-a-time by fitting a sequence of separate models for 3-month MCID in pain severity (modified Poisson regression) and for any follow-up (logistic regression). Those baseline covariates that, when included in models with study arm and the base set of adjustment variables, reached statistical significance at the 0.10 level in model for MCID and in the model for any follow-up (**Supplement 5**) were added to the weight model.

The final model for the non-response weights took the following form:

$$\text{logit}(E(R_i|X_i, \mathbf{Z}_i)) = \beta_0 + \beta_{PT}I(X_i = PT) + \beta_{HC}I(X_i = HC) + \beta_z\mathbf{Z}_i,$$

where R_i indicates having at least one observed follow-up measurement. Predictions, p_i , from this model are used to calculate weights $w_i = \frac{1}{p_i}$. The w_i were then scaled to the study sample size and included as weights in the GEE

estimation model described in the imputation section (step 2). The fitted weight model is presented in **eTable 5**, and **eTable 6** and **eFigure 1** provide summaries of the predicted probabilities and corresponding weights.

eTable 1. Primary outcome follow-up rates and patterns overall and by arm.

	OVERALL		Usual Care plus		painTRAINER		Health Coach	
	N	%	N	%	N	%	N	%
Randomized	2331		777		776		778	
Follow-up								
3 months	1798	77.1	621	79.9	542	69.8	635	81.6
6 months	1790	76.8	622	80.1	547	70.5	621	79.8
12 months	1861	79.8	639	82.2	583	75.1	639	82.1
At least 1 follow-up	2036	87.3	703	90.5	643	82.9	690	88.7
No follow-up	295	12.7	74	9.5	133	17.1	88	11.3
One observed follow-up	188	8.1	64	8.2	77	9.9	47	6.0
Two observed follow-ups	283	12.1	99	12.7	103	13.3	81	10.4
Three observed follow-ups	1568	67.3	540	69.5	464	59.8	564	72.5

eTable 2. Baseline variables used in process of selecting covariates related to missingness and to MCID of pain severity at 3 months.

Category	Variables
Variables included in all models	Study arm, sex (male vs. female), age (≥ 65 vs. < 65 years), clinical site, rural/medically underserved residency (urban vs. rural), multisite pain (> 1 site of pain), and co-occurring mental health conditions (presence of depression and/or anxiety diagnoses)
VARIABLES PROVIDED FOR SELECTION	
Demographic characteristics	Race and ethnicity (Asian, Black, Hispanic, Multiple, Native Hawaiian or Pacific Islander, White non-Hispanic), education, unemployment, household income ($< 25K$, $25- < 50K$, $50- < 100K$, $100- < 150K$, $150K$ or more), married/partnered (yes vs. no), substance abuse disorder diagnosis (present vs. absent)
Pain Related & expectations	Comorbidities (Charlson, EHR), number of musculoskeletal pain conditions in the past year (EHR), number of pain-related healthcare encounters (EHR), presence of specific types of pain (back, neck, arm/hand/wrist, legs/feet, head, abdomen/pelvis, tooth/jaw, general pain), pain duration
Clinical Characteristics/Health Habits	Depression (PHQ-8), anxiety (GAD-7), PROMIS sleep disturbance, presence of negative social determinants of health – individual, presence of any, and count (financial strain, food insecurity, housing instability, and/or transportation insecurity)
Baseline secondary and Tertiary Outcomes	Pain intensity, pain-related interference, PROMIS social role functioning, PROMIS physical functioning, PEG score
FINAL VARIABLES SELECTED	
Additional variables selected for inclusion in imputation and estimation models	Education (any college vs. HS degree or less), unemployment (unemployed vs. employed [full or part-time])
Additional variables selected for inclusion in non-response weight model.	PEG score, Depression (PHQ-8 score), Anxiety (GAD-7 score), presence of any negative social determinant of health, education (any college vs. HS degree or less)

eTable 3. Imputation step 1: Summary of fitted imputation model for primary analysis.

Step 1. Imputation Model (Linear)				
Outcome: Change in pain severity		Overall	N	
Number of observations used		6108	2036	
Number of missing values		659		
Variable	Est.	95% CI		p-value
Intercept	-1.24	-1.42	-1.06	<.001
painTRAINER (vs. Usual Care plus)	0.49	0.39	0.59	<.001
Health Coach (vs. Usual Care plus)	0.45	0.36	0.54	<.001
Time: 6M	0.13	0.07	0.19	0.025
Time: 12M	0.31	0.24	0.38	<.001
Pattern: 2 observed follow-ups	-0.28	-0.47	-0.08	0.150
Pattern: 1 observed follow-up	-0.20	-0.36	-0.03	0.226
painTRAINER Time 6M	0.02	-0.07	0.11	0.853
painTRAINER Time 12M	0.01	-0.09	0.12	0.916
Health Coach Time 6M	0.08	-0.01	0.17	0.351
Health Coach Time 12M	-0.06	-0.16	0.04	0.535
Pattern 2 painTRAINER	-0.30	-0.58	-0.02	0.280
Pattern 1 painTRAINER	-0.21	-0.45	0.02	0.368
Pattern 2 Health Coach	-0.47	-0.74	-0.19	0.090
Pattern 1 Health Coach	0.18	-0.07	0.44	0.468
Sex: Male	-0.22	-0.29	-0.14	0.003
Age >=65	0.19	0.12	0.27	0.013
Site 2	0.07	-0.03	0.17	0.479
Site 3	0.19	0.08	0.29	0.069
Site 4	0.15	0.05	0.24	0.124
Residence:Urban	0.07	0.00	0.15	0.306
1+ musculoskeletal pain conditions	-0.22	-0.30	-0.14	0.004
No Mental health diagnosis EHR	0.26	0.19	0.33	<.001
Pain severity	0.36	0.33	0.38	<.001
Education:AA degree (ref = High school or less)	-0.02	-0.11	0.08	0.866
Education:AA degree (ref = High school or less)	0.16	0.07	0.25	0.077
Education:Graduate degree (ref = High school or less)	0.39	0.29	0.49	<.001
Not employed	-0.55	-0.62	-0.47	<.001

eTable 4. Imputation step 2: Summary of primary estimation model fit to imputed data with weighting.

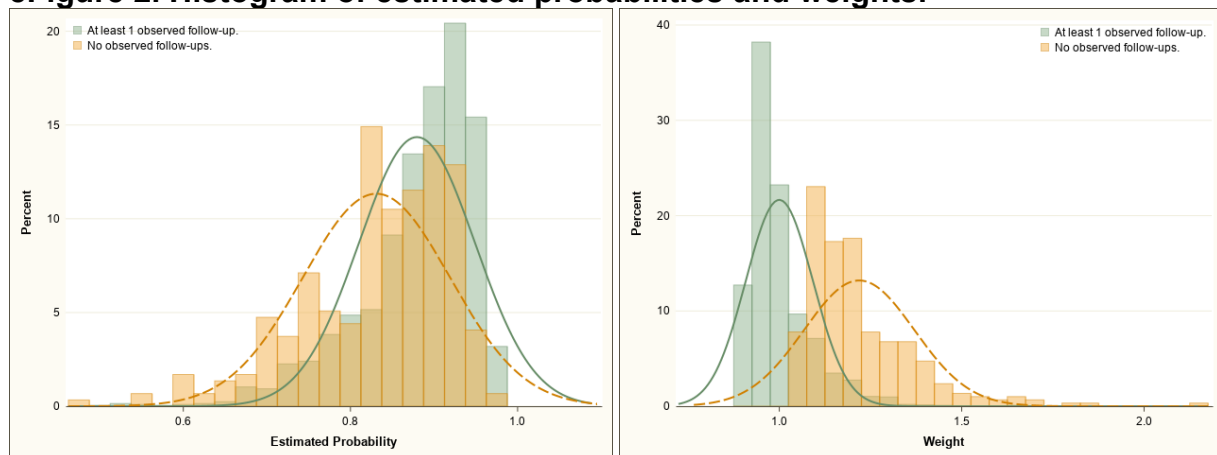
Step 2. Estimation Model (Poisson) – Imputation with Weights. Outcome: Binary 30% improvement in pain severity.	Overall	Persons		
Number of Observations Used	6108	2036		
Sum of Weights	6108			
Variable	RR	95% CI		p-value
painTRAINER (vs. Usual Care plus)	1.28	1.06	1.55	0.010
Health Coach (vs. Usual care plus)	1.54	1.3	1.82	<.001
Health Coach vs. painTRAINER 3M	1.2	1.03	1.4	0.019
painTRAINER vs. Usual Care plus 6M	1.44	1.21	1.7	<0.001
painTRAINER vs. Usual Care plus 12M	1.32	1.13	1.54	<0.001
Health Coach vs. Usual Care plus 6M	1.62	1.39	1.9	<0.001
Health Coach vs. Usual Care plus 12M	1.41	1.25	1.59	<0.001
Health Coach vs. painTRAINER 3M	1.2	1.03	1.4	0.019
Health Coach vs. painTRAINER 6M	1.13	0.98	1.3	0.086
Health Coach vs. painTRAINER 12M	1.07	0.96	1.19	0.215
Sex: Male (vs. Female)	0.8	0.71	0.91	<0.001
Age >=65 (vs. <65)	1.15	1	1.33	0.046
Site	1.07	0.89	1.28	0.498
Site	1.15	0.97	1.37	0.102
Site	1.17	1.02	1.34	0.028
Residence: Urban (vs. Rural)	1.07	0.96	1.19	0.226
1+ musculoskeletal pain conditions (vs. 1)	0.87	0.78	0.97	0.014
Mental health diagnosis EHR (vs. None)	1.15	1.04	1.27	0.008
Pain severity	0.99	0.96	1.02	0.451
Education: AA degree (vs. HS or less)	0.97	0.82	1.16	0.770
Education: AA degree (ref vs. HS or less)	1.16	1	1.33	0.046
Education: Graduate degree (vs. HS or less)	1.35	1.13	1.61	<0.001
Not employed (vs. Employed)	0.72	0.64	0.8	<0.001

eTable 5. Summary of logistic weighting model for any observed follow-up.

Variable	OR*	95% CI		P-value
Usual Care plus (vs. painTRAINER)	2.05	1.50	2.79	<0.001
Health Coach (vs. painTRAINER)	1.68	1.25	2.26	<0.001
Male Sex	0.76	0.57	1.00	0.053
Age >=65	0.97	0.70	1.35	0.859
Site	2.93	1.97	4.36	<0.001
Site	3.43	2.27	5.18	<0.001
Site	2.48	1.70	3.61	<0.001
Residence Urban	0.67	0.49	0.90	0.009
>1 Musculoskeletal Pain Condition	0.97	0.72	1.29	0.812
No Mental Health Diagnosis	1.21	0.92	1.59	0.165
Pain Severity	1.05	0.93	1.18	0.417
Not employed	1.49	1.11	2.00	0.008
No Negative Social Determinant of Health	1.34	1.01	1.77	0.042
PEG score	0.96	0.92	1.00	0.037
PHQ-8	0.97	0.94	1.01	0.125
GAD-7	0.99	0.96	1.02	0.524
c-statistic	0.679			
* OR = Odds Ratio				
Health care site information has been excluded.				

eTable 6. Summary of estimated weights.

	Sum	Mean	SD	Min.	P25	P50	P75	Max.
Estimated probability of having at least 1 follow-up.	--	0.88	0.07	0.50	0.85	0.90	0.93	0.98
Calculated weight used for analysis.	2036	1.00	0.09	0.89	0.94	0.97	1.03	1.75

eFigure 2. Histogram of estimated probabilities and weights.

SENSITIVITY ANALYSES

Missing Data Analyses: Sensitivity analysis results for different missing data assumptions for the MCID on pain severity primary outcome are presented in **eTable 7** below. **eTable 7** provides results for different models assumptions for missing data that were pre-specified in the SAP including: a) complete-case analysis (missing outcomes removed) adjusting for baseline pain severity, age, sex, clinical site, rural/medically underserved residency, multisite pain, and co-occurring mental health conditions; b) worst case/best case analysis (missing outcomes for participants randomized to PT or HC are assumed 0 (worst case) and missing outcomes for participants randomized to UC are 1 (best case)) adjusting for baseline pain severity, age, sex, clinical site, rural/medically underserved residency, multisite pain, and co-occurring mental health conditions.

eTable 7. Pre-specified Sensitivity Analyses for different approaches to handling missing data.

	Adjusted % with a 30% improvement in pain score, % (95% CI) ^b			Adjusted Relative Risk (95% CI) ^b			P val ^c	Adjusted Number Needed to Treat (95% CI) ^{d,e}		
	painTRAINER (PT)	Health Coach (HC)	Usual Care Plus (UC)	PT vs. UC	HC vs. UC	HC vs. PT		PT vs. UC	HC vs. UC	HC vs. PT
Complete-Case analysis: Adjusted GEE without imputation or weighting (n=1,798 at 3-months; UC=621, PT=542, HC=635)										
3 months ^a	32.2 (28.5, 36.3)	35.5 (32.0, 39.3)	24.0 (20.9, 27.6)	1.34 (1.12, 1.61)	1.48 (1.24, 1.75)	1.10 (0.94, 1.29)	<0.001	13 (8, 33)	9 (7, 15)	31 (13, -60)
6 months	40.1 (36.3, 44.3)	41.1 (37.4, 45.1)	26.6 (23.4, 30.3)	1.51 (1.28, 1.77)	1.54 (1.32, 1.81)	1.02 (0.90, 1.17)	<0.001	8 (6, 13)	7 (6, 11)	101 (16, -22)
12 months	41.0 (37.3, 45.1)	41.5 (37.9, 45.4)	30.5 (27.2, 34.3)	1.34 (1.16, 1.56)	1.36 (1.17, 1.57)	1.01 (0.89, 1.15)	<0.001	10 (7, 20)	10 (7, 15)	217 (22, -28)
Worst-Case/Best-Case: Adjusted GEE assuming missing outcome data is Worst Case/Best Case (n=2,331; UC=777, PT=776, HC=778)										
3 months ^a	22.7 (20.0, 25.8)	29.3 (26.3, 32.6)	39.4 (36.1, 43.0)	0.58 (0.49, 0.67)	0.74 (0.65, 0.85)	1.29 (1.09, 1.52)	<0.001	-6 (-9, -5)	-10 (-18, -7)	16 (10, 40)
6 months	28.6 (25.6, 31.9)	33.1 (30.0, 36.5)	41.3 (38.0, 44.9)	0.69 (0.60, 0.79)	0.80 (0.70, 0.91)	1.16 (1.00, 1.34)	<0.001	-8 (-13, -6)	-13 (-26, -8)	23 (12, 227)
12 months	31.3 (28.2, 34.6)	34.4 (31.2, 37.8)	43.0 (39.6, 46.7)	0.73 (0.64, 0.83)	0.80 (0.70, 0.91)	1.10 (0.96, 1.26)	<0.001	-9 (-15, -7)	-12 (-21, -8)	33 (15, -194)
^a Primary timepoint										
^b Adjusted mean % and adjusted RR were calculated from a modified Poisson regression model fit using generalized estimating equations (GEE) for each binary outcome. Adjusted mean % assumes the mean of the covariate distribution to calculate randomized population average effects. Adjustment variables: Sex, age, site, rural vs. urban residence, more than 1 musculoskeletal pain condition, presence of mental health diagnoses, and pain severity at baseline.										
^c Omnibus P value is the Wald Test to assess if there is any difference between the three groups. To control for multiple compairsons between-group comparisons should only be compared if the Omnibus P value<0.05 following the least significant difference approach.										
^d Adjusted Number needed to treat (NNT) to gain one additional 30% improvement, i.e., one over the difference in the adjusted percent with an 30% improvement or greater.										
^e If CIs for the absolute risk difference contain zero the NNT confidence limits appear non-sensical due to division by zero. The NNT estimate is interpreted as being consistent with a NNT from the lower limit (positive number) to infinity, and a number needed to harm (NNH) from the upper limit (negative number) to negative infinity.										

Protocol Change Subset Analyses: Sensitivity analysis results for different protocol changes for the primary outcome MCID on pain severity are presented in **eTable 8**. The results in **eTable 8** shows the results of the primary analysis restricting to subsets of the randomized population after the following protocol changes were made: a) enhancement to the painTRAINER engagement activities (restricts to those randomized after 8/9/2021) and b) reminder letter sent to all participants prior to follow-up survey reach out (restrict to those randomized after 3/31/2022). Specifically, to improve engagement in the painTRAINER intervention the study team added motivational interviewing-based language to the onboarding script for those randomized to the online program. This change was implemented on 8/9/2021 after 454 patients were enrolled in the study. This sensitivity analysis was pre-specified in the SAP to evaluate whether the “enhanced” painTRAINER engagement activities improved the effect of the painTRAINER intervention. The follow-up letter analysis was not pre-specified in the SAP, but we have added this post-hoc analysis after observing large improvements in follow-up rates (82.7% overall at 3 months and 90.1% with at least one follow-up measure) after the study added a reminder letter, mailed to all participants before the due date of their follow-up survey, to remind them to expect an email or a call (depending on participant preference) from the study team, and encouraged participants to complete the survey when contacted.

eTable 8. Results of a Subset of Participants after two different protocol changes: Enhancement to painTRAINER (randomized after 8/9/2021) and Adding follow-up letters to improve follow-up rates (randomized after 3/31/2022)

	Adjusted % with a 30% improvement in pain score, % (95% CI) ^b			Adjusted Relative Risk ^c (95% CI) ^b			P val ^c	Adjusted Number Needed to Treat (95% CI) ^{d,e}		
	painTRAINER (PT)	Health Coach (HC)	Usual Care Plus (UC)	PT vs. UC	HC vs. UC	HC vs. PT		PT vs. UC	HC vs. UC	HC vs. PT
Randomized prior to 8/9/2021, N=454; 366 with at least 1 follow-up; PT=149, HC=153, UC=152 at 3-months										
3 months ^a	20.5 (14.2, 29.6)	28.5 (23.4, 34.8)	21.3 (15.2, 29.8)	0.96 (0.59, 1.57)	1.34 (0.92, 1.95)	1.39 (0.92, 2.10)	0.138	-120 (11, -10)	14 (7, -67)	13 (6, -84)
6 months	32.9 (25.1, 43.0)	32.4 (27.8, 37.7)	25.2 (18.5, 34.2)	1.31 (0.88, 1.95)	1.29 (0.92, 1.80)	0.98 (0.73, 1.33)	0.313	13 (6, -27)	14 (7, -56)	-185 (11, -10)
12 months	32.6 (25.0, 42.5)	37.7 (32.2, 44.3)	20.5 (14.7, 28.6)	1.59 (1.05, 2.41)	1.84 (1.29, 2.63)	1.16 (0.85, 1.57)	0.003	9 (5, 75)	6 (4, 12)	20 (7, -20)
Enhancement to Pain Trainer (Randomized after 8/9/2021, N=1877; 1670 with at least 1 follow-up; UC=512, PT=448, HC=527 at 3-months)										
3 months ^a	28.0 (24.4, 32.0)	32.7 (29.5, 36.2)	20.4 (17.4, 23.9)	1.37 (1.11, 1.69)	1.60 (1.33, 1.93)	1.17 (0.99, 1.38)	<0.001	14 (8, 39)	9 (6, 14)	22 (11, -327)
6 months	33.8 (30.1, 38.1)	38.2 (34.5, 42.3)	22.2 (19.1, 25.7)	1.53 (1.27, 1.84)	1.72 (1.44, 2.06)	1.13 (0.97, 1.32)	<0.001	9 (6, 16)	7 (5, 10)	24 (11, -88)
12 months	36.9 (33.1, 41.2)	38.1 (36.0, 40.3)	28.6 (25.2, 32.5)	1.29 (1.09, 1.52)	1.33 (1.16, 1.52)	1.03 (0.92, 1.16)	<0.001	13 (8, 34)	11 (8, 19)	86 (18, -31)
Reminder Letters Added (Randomized after 3/31/2022, N=1038; 935 with at least 1 follow-up; UC=296, PT=262, HC=300 at 3-months)										
3 months ^a	31.6 (26.7, 37.3)	34.1 (29.5, 39.5)	20.5 (16.6, 25.3)	1.54 (1.18, 2.01)	1.67 (1.30, 2.14)	1.08 (0.87, 1.34)	<0.001	9 (6, 23)	8 (5, 14)	40 (11, -23)
6 months	35.6 (30.6, 41.4)	40.6 (35.8, 46.1)	21.3 (17.5, 26.1)	1.67 (1.30, 2.14)	1.90 (1.50, 2.41)	1.14 (0.94, 1.39)	<0.001	8 (5, 14)	6 (4, 8)	20 (9, -43)
12 months	38.2 (33.1, 44.0)	38.9 (35.6, 42.5)	26.1 (21.9, 31.2)	1.46 (1.17, 1.83)	1.49 (1.23, 1.80)	1.02 (0.87, 1.20)	<0.001	9 (6, 21)	8 (6, 14)	133 (15, -19)
^a Primary timepoint										
^b Adjusted mean % and adjusted RR were calculated from a modified Poisson regression model fit using generalized estimating equations (GEE) for each binary outcome. Adjusted mean % assumes the mean of the covariate distribution to calculate randomized population average effects. Adjustment variables: Sex, age, site, rural vs. urban residence, more than 1 musculoskeletal pain condition, presence of mental health diagnoses, and pain severity at baseline.										
^c Omnibus P value is the Wald Test to assess if there is any difference between the three groups. To control for multiple comparisons between-group comparisons should only be compared if the Omnibus P value<0.05 following the least significant difference approach.										
^d Number needed to treat (NNT) to gain one additional 30% improvement, i.e., one over the difference in the adjusted percent with an 30% improvement or greater.										
^e When CIs for the absolute risk difference contain zero the NNT confidence limits appear non-sensical due to division by zero. The NNT estimate is interpreted as being consistent with a NNT from the lower limit (positive number) to infinity, and a number needed to harm (NNH) from the upper limit (negative number) to negative infinity.										

ADDITIONAL UNADJUSTED DATA AND ANALYSES

For completeness we replicated **Tables 2** and **3** in the main manuscript but present unadjusted analyses with no adjustment, weighting or imputation for all primary and secondary outcomes. **eTable 9** presents unadjusted means of the binary primary and secondary outcomes. For Table 3 unadjusted results, we present two tables in which **eTable 10** provides unadjusted mean change from baseline secondary outcomes and **eTable 11** provides unadjusted mean secondary outcomes.

eTable 9. Unadjusted percentage of participants with clinically meaningful improvement by treatment group and adjusted relative risk between group comparisons for the binary primary outcome (pain severity) and secondary pain outcomes

	Unadjusted Number with 30% improvement in pain score, n/N (%)			Relative Risk (95% CI) ^b		
	painTRAINER (PT)	Health Coach (HC)	Usual Care plus (UC)	PT vs UC	HC vs. UC	HC vs. PT
Pain Severity						
3 months ^a	179/542 (33.0)	229/635 (36.1)	151/621 (24.3)	1.36 (1.13, 1.63)	1.48 (1.26, 1.75)	1.09 (0.94, 1.27)
6 months	225/547 (41.1)	259/621 (41.7)	167/622 (26.8)	1.53 (1.30, 1.80)	1.55 (1.32, 1.83)	1.01 (0.88, 1.17)
12 months	246/583 (42.2)	269/639 (42.1)	197/639 (30.8)	1.37 (1.18, 1.59)	1.37 (1.21, 1.54)	1.00 (0.90, 1.10)
Pain Intensity						
3 months ^a	164/543 (30.2)	192/635 (30.2)	129/621 (20.8)	1.46 (1.19, 1.78)	1.46 (1.18, 1.79)	1.00 (0.83, 1.21)
6 months	183/548 (33.4)	222/622 (35.7)	153/623 (24.6)	1.36 (1.13, 1.63)	1.45 (1.20, 1.74)	1.06 (0.90, 1.26)
12 months	218/583 (37.4)	238/641 (37.1)	191/639 (29.9)	1.25 (1.07, 1.47)	1.24 (1.08, 1.42)	0.99 (0.87, 1.12)
Pain-related Interference						
3 months ^a	206/543 (37.9)	249/635 (39.2)	169/620 (27.3)	1.39 (1.17, 1.64)	1.44 (1.25, 1.65)	1.03 (0.92, 1.17)
6 months	249/546 (45.6)	277/620 (44.7)	176/621 (28.3)	1.61 (1.38, 1.88)	1.58 (1.35, 1.85)	0.98 (0.86, 1.12)
12 months	267/583 (45.8)	293/638 (45.9)	221/638 (34.6)	1.32 (1.15, 1.52)	1.33 (1.18, 1.48)	1.00 (0.91, 1.10)
^a Primary timepoint						
^b Relative Risk and 95% Confidence Intervals were calculated from a modified Poisson regression model fit using generalized estimating equations (GEE) for each binary outcome without adjustment.						

eTable 10. Unadjusted Median and Interquartile Ranges for change from baseline of continuous secondary outcomes by treatment group and unadjusted mean differences between group comparisons

	Change from Baseline Median (25%-tile, 75%tile)			Unadjusted Between-Group Mean Differences (95% CI) ^b			Omnibus <i>P</i> value
	painTRAINER (PT)	Health Coach (HC)	Usual Care plus (UC)	PT vs. UC	HC vs. UC	HC vs. PT	
Pain Severity							
3 months ^a	-1.1 (-2.1, -0.1)	-1.1 (-2.3, -0.1)	-0.6 (-1.6, 0.2)	-0.4 (-0.6, -0.3)	-0.4 (-0.6, -0.3)	0.0 (-0.2, 0.2)	<0.001
6 months	-1.3 (-2.5, -0.1)	-1.4 (-2.5, -0.2)	-0.8 (-1.8, 0.2)	-0.4 (-0.7, -0.2)	-0.5 (-0.8, -0.3)	-0.1 (-0.3, 0.1)	<0.001
12 months	-1.3 (-2.5, -0.3)	-1.4 (-2.5, -0.2)	-0.8 (-2.2, 0.1)	-0.4 (-0.7, -0.2)	-0.4 (-0.6, -0.2)	0.0 (-0.1, 0.2)	<0.001
Pain Intensity							
3 months ^a	-0.8 (-1.8, 0.0)	-0.8 (-2.0, 0.0)	-0.5 (-1.5, 0.2)	-0.3 (-0.5, -0.2)	-0.3 (-0.5, -0.1)	0.0 (-0.1, 0.2)	<0.001
6 months	-1.0 (-2.0, 0.0)	-1.0 (-2.2, 0.0)	-0.8 (-1.5, 0.2)	-0.3 (-0.5, -0.1)	-0.3 (-0.5, -0.2)	-0.1 (-0.3, 0.1)	<0.001
12 months	-1.0 (-2.2, 0.0)	-1.0 (-2.2, 0.0)	-0.8 (-2.0, 0.2)	-0.3 (-0.5, -0.1)	-0.2 (-0.4, -0.1)	0.0 (-0.1, 0.2)	<0.001
Pain-related Interference							
3 months ^a	-1.3 (-2.4, -0.1)	-1.3 (-2.6, 0.0)	-0.7 (-1.9, 0.1)	-0.5 (-0.7, -0.3)	-0.5 (-0.7, -0.3)	0.0 (-0.2, 0.2)	<0.001
6 months	-1.4 (-2.9, 0.0)	-1.6 (-2.9, -0.1)	-0.9 (-2.1, 0.1)	-0.6 (-0.8, -0.3)	-0.6 (-0.9, -0.4)	-0.1 (-0.4, 0.2)	<0.001
12 months	-1.6 (-3.0, -0.1)	-1.6 (-3.0, -0.1)	-1.0 (-2.6, 0.1)	-0.5 (-0.8, -0.3)	-0.5 (-0.7, -0.3)	0.0 (-0.2, 0.3)	<0.001
Social Role Functioning							
3 months ^a	1.8 (-2.0, 5.8)	1.9 (-1.9, 6.0)	0.0 (-2.2, 4.3)	0.9 (0.1, 1.7)	1.5 (0.7, 2.2)	0.5 (-0.2, 1.3)	<0.001
6 months	1.9 (-1.9, 6.5)	1.9 (-1.9, 7.4)	1.5 (-3.2, 5.5)	1.3 (0.4, 2.2)	1.7 (0.9, 2.6)	0.4 (-0.5, 1.4)	<0.001
12 months	1.9 (-1.9, 7.5)	2.0 (-1.7, 7.7)	0.0 (-3.2, 5.8)	1.4 (0.4, 2.3)	1.4 (0.8, 2.1)	0.1 (-0.7, 0.8)	<0.001
Physical Functioning							
3 months ^a	1.6 (-0.8, 3.6)	1.2 (-0.8, 3.8)	0.9 (-1.4, 3.2)	0.6 (0.1, 1.1)	0.8 (0.3, 1.2)	0.1 (-0.3, 0.6)	0.002
6 months	1.7 (-0.8, 4.4)	1.2 (-0.9, 4.4)	1.0 (-1.0, 3.6)	0.8 (0.3, 1.4)	0.5 (0.1, 0.9)	-0.3 (-0.8, 0.2)	<0.001
12 months	1.9 (-0.9, 5.0)	1.3 (-1.2, 4.6)	0.9 (-1.7, 3.6)	1.1 (0.5, 1.7)	0.8 (0.2, 1.5)	-0.3 (-0.9, 0.4)	<0.001
^a Primary timepoint							
^b Mean change from baseline and unadjusted between group mean difference confidence intervals were calculated using linear regression model fit using generalized estimating equations (GEE) for each continuous outcome.							

eTable 11. Unadjusted Median and Interquartile Ranges of continuous secondary outcomes by treatment group and unadjusted mean differences in between group comparisons

	Raw Score Median (25%-tile, 75%tile)			Unadjusted Between-Group Mean Differences (95% CI) ^b			Omnibus <i>P</i> value
	painTRAINER (PT)	Health Coach (HC)	Usual Care plus (UC)	PT vs. UC	HC vs. UC	HC vs. PT	
Pain Severity							
3 months ^a	5.1 (4.9, 5.2)	4.6 (4.4, 4.7)	4.6 (4.5, 4.7)	-0.5 (-0.7, -0.3)	-0.5 (-0.7, -0.3)	0.0 (-0.2, 0.2)	<0.001
6 months	5.0 (4.8, 5.1)	4.4 (4.2, 4.6)	4.4 (4.2, 4.5)	-0.6 (-0.8, -0.3)	-0.6 (-0.8, -0.3)	-0.0 (-0.2, 0.2)	<0.001
12 months	4.8 (4.6, 4.9)	4.2 (4.1, 4.4)	4.3 (4.2, 4.5)	-0.5 (-0.8, -0.3)	-0.4 (-0.6, -0.2)	0.1 (-0.1, 0.3)	<0.001
Pain Intensity							
3 months ^a	4.9 (4.7, 5.0)	4.5 (4.3, 4.6)	4.5 (4.4, 4.6)	-0.4 (-0.6, -0.2)	-0.3 (-0.5, -0.1)	0.0 (-0.2, 0.2)	<0.001
6 months	4.8 (4.6, 4.9)	4.4 (4.2, 4.5)	4.4 (4.3, 4.5)	-0.4 (-0.6, -0.2)	-0.4 (-0.6, -0.2)	-0.0 (-0.2, 0.2)	<0.001
12 months	4.6 (4.4, 4.7)	4.2 (4.1, 4.4)	4.3 (4.2, 4.4)	-0.3 (-0.6, -0.1)	-0.2 (-0.4, -0.1)	0.1 (-0.1, 0.3)	<0.001
Pain-related Interference							
3 months ^a	5.2 (5.0, 5.3)	4.6 (4.4, 4.8)	4.6 (4.5, 4.7)	-0.6 (-0.8, -0.3)	-0.6 (-0.8, -0.4)	0.0 (-0.2, 0.2)	<0.001
6 months	5.1 (4.9, 5.2)	4.4 (4.2, 4.6)	4.4 (4.2, 4.6)	-0.7 (-0.9, -0.4)	-0.7 (-0.9, -0.4)	-0.0 (-0.3, 0.2)	<0.001
12 months	4.9 (4.7, 5.1)	4.2 (4.0, 4.4)	4.3 (4.2, 4.5)	-0.6 (-0.9, -0.4)	-0.5 (-0.8, -0.3)	0.1 (-0.1, 0.3)	<0.001
Social Role Functioning							
3 months ^a	43.4 (42.8, 44.0)	44.6 (43.9, 45.3)	44.8 (44.4, 45.2)	1.2 (0.3, 2.1)	1.4 (0.7, 2.2)	0.2 (-0.5, 1.0)	0.001
6 months	43.6 (42.9, 44.2)	45.4 (44.7, 46.0)	45.1 (44.7, 45.4)	1.8 (0.9, 2.7)	1.5 (0.8, 2.2)	-0.3 (-1.0, 0.4)	<0.001
12 months	43.9 (43.3, 44.6)	45.8 (45.1, 46.5)	45.3 (44.8, 45.8)	1.8 (0.9, 2.8)	1.4 (0.5, 2.2)	-0.5 (-1.3, 0.3)	<0.001
Physical Functioning							
3 months ^a	37.7 (37.1, 38.2)	38.2 (37.7, 38.8)	38.2 (37.7, 38.8)	0.6 (-0.2, 1.3)	0.6 (-0.2, 1.3)	0.0 (-0.8, 0.8)	0.212
6 months	37.9 (37.3, 38.4)	38.9 (38.4, 39.5)	38.3 (37.8, 38.8)	1.1 (0.3, 1.9)	0.4 (-0.3, 1.1)	-0.6 (-1.4, 0.1)	0.005
12 months	37.9 (37.3, 38.4)	39.1 (38.5, 39.7)	38.8 (38.1, 39.4)	1.2 (0.4, 2.0)	0.9 (0.0, 1.8)	-0.3 (-1.2, 0.6)	0.001
Patient Global Impression of Change – Pain							
3 months ^a	2.9 (2.8, 2.9)	2.0 (1.0, 3.0)	1.9 (1.9, 2.0)	-0.8 (-0.9, -0.6)	-0.9 (-1.0, -0.8)	-0.2 (-0.3, -0.0)	<0.001
6 months	2.8 (2.7, 2.9)	2.1 (2.0, 2.2)	2.1 (2.0, 2.2)	-0.7 (-0.8, -0.5)	-0.7 (-0.8, -0.5)	-0.0 (-0.2, 0.1)	<0.001
12 months	2.8 (2.7, 2.9)	2.1 (2.0, 2.2)	2.2 (2.1, 2.3)	-0.7 (-0.8, -0.5)	-0.6 (-0.7, -0.4)	0.1 (-0.1, 0.2)	<0.001
Patient Global Impression of Change – General							
3 months ^a	2.7 (2.6, 2.7)	2.0 (1.0, 2.0)	1.6 (1.5, 1.6)	-0.8 (-0.9, -0.6)	-1.1 (-1.2, -1.0)	-0.3 (-0.5, -0.2)	<0.001
6 months	2.5 (2.4, 2.6)	2.0 (1.9, 2.1)	1.8 (1.7, 1.9)	-0.6 (-0.7, -0.4)	-0.8 (-0.9, -0.6)	-0.2 (-0.3, -0.0)	<0.001
12 months	2.6 (2.4, 2.7)	1.9 (1.8, 2.1)	1.9 (1.7, 2.0)	-0.6 (-0.8, -0.4)	-0.7 (-0.8, -0.5)	-0.1 (-0.2, 0.1)	<0.001

^a Primary timepoint; ^b Unadjusted between group mean differences and 95% confidence intervals were calculated using linear regression model fit using generalized estimating equations (GEE) for each continuous outcome.

eREFERENCES

1. Zou G. A modified poisson regression approach to prospective studies with binary data. *Am J Epidemiol.* Apr 1 2004;159(7):702-6.
2. Wang M, Fitzmaurice GM. A simple imputation method for longitudinal studies with non-ignorable non-responses. *Biom J.* Apr 2006;48(2):302-18.
3. White IR, Thompson SG. Adjusting for partially missing baseline measurements in randomized trials. *Stat Med.* Apr 15 2005;24(7):993-1007.